

Pre-Registered Predictions for the Energy-Flow Cosmology Model Against the Rubin Observatory LSST DP2 Cosmic Shear Dataset

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This document seals predictions prior to data availability.

Abstract. We present frozen, pre-registered predictions of the Energy-Flow Cosmology (EFC) model for the cosmic shear two-point correlation functions $\xi_{\pm}(\theta)$ and the convergence power spectrum $C_{\ell}^{\kappa\kappa}$ expected from the Rubin Observatory LSST Data Preview 2 (DP2) dataset. The central prediction is a $\sim 10\%$ enhancement in the lensing amplitude relative to Λ CDM at angular scales $\theta \gtrsim 20'$ ($\ell \lesssim 300$), driven by the EFC modified gravity parameter $\eta \approx 1.10$ (the gravitational slip $\eta \equiv \Phi/\Psi$, convention fixed throughout). This enhancement shifts the effective S_8 parameter upward by $\Delta S_8 \approx +0.015$ relative to the Λ CDM prediction, providing a clean discriminant against Λ CDM. A falsification criterion is specified: if the observed $\xi_+(\theta)$ at $\theta \in [20', 300']$ lies below the EFC band at $> 2\sigma$, the KC4 lensing constraint is ruled out. All numerical values are derived from independently archived sources and sealed by SHA-256 hash.

1 Introduction and Motivation

This document constitutes a formal pre-registration of EFC model predictions against the Rubin Observatory LSST DP2 cosmic shear dataset. It is a companion to the sealed EFC pre-registration document (DOI: [10.6084/m9.figshare.32010399](https://doi.org/10.6084/m9.figshare.32010399)), which covers BAO, RSD, and growth observables. The present document targets the weak-lensing sector specifically.

The EFC model modifies the gravitational potentials at late times through a regime-activated response in the μ - Σ eigenmode. In the modified gravity parameterisation, $\mu(a, k)$ governs the clustering potential and $\Sigma(a, k)$ governs the lensing potential. Their ratio $\eta \equiv \Phi/\Psi$ (the gravitational slip) is the key observable targeted by cosmic shear. The EFC prediction $\eta \approx 1.10$ (KC4 constraint) implies that lensing is enhanced relative to the growth of structure, making cosmic shear the natural test of this prediction.

1.0 Why Rubin DP2?

The Rubin LSST DP2 dataset will cover $\sim 300 \text{ deg}^2$ to full survey depth, providing the first Stage-IV quality cosmic shear measurement. The expected source galaxy number density ($n_{\text{eff}} \gtrsim 20 \text{ arcmin}^{-2}$) and photometric redshift quality make DP2 sensitive to the angular scales ($20' \lesssim \theta \lesssim 300'$) where the EFC enhancement is largest.

2 Theoretical Framework

2.1 EFC Modified Gravity Parameterisation

The EFC model is fully specified by the Boltzmann-level parameterisation archived at DOI [31876324](https://doi.org/10.6084/m9.figshare.32013738). For the lensing sector, the relevant modifications are:

$$\mu(a, k) = 1 + \alpha_\mu \mathcal{G}(a), \quad (1)$$

$$\Sigma(a, k) = 1 + \alpha_\Sigma \mathcal{G}(a), \quad (2)$$

where $\mathcal{G}(a)$ is the EFC regime gate function that activates at $a \gtrsim a_t = 0.3$ (corresponding to $z \lesssim 2.3$), and $\alpha_\mu, \alpha_\Sigma$ are fixed parameters derived from the MCMC posterior sealed in the parent pre-registration.

The gravitational slip parameter follows from Eqs. (1)–(2). We adopt the convention $\eta \equiv \Phi/\Psi$ throughout this document, where Φ is the lensing (Weyl) potential and Ψ the Newtonian potential:

$$\eta(a) \equiv \frac{\Phi(a)}{\Psi(a)} = \frac{1 + \alpha_\Sigma \mathcal{G}(a)}{1 + \alpha_\mu \mathcal{G}(a)}. \quad (3)$$

2.2 Frozen Parameter Values (KC4)

The following parameter values are frozen at the KC4 constraint level, derived from the NUTS posterior (sealed hash `7a850cfa58477701...`, 2026-02-18):

Table 1: Frozen EFC lensing parameters (KC4 level). No modifications permitted after pre-registration.

Parameter	Value	Uncertainty	Source
$\eta(z=0)$	1.10	± 0.05	KC4 posterior (NUTS)
α (EFC)	-0.695	± 0.02	Mean of two sealed runs
a_t (regime transition)	0.30	fixed	Ledger v3.11
Δ (transition width)	0.30	fixed	Ledger v3.11
σ_8^{EFC}	0.833	± 0.010	Boltzmann benchmark
Ω_m	0.310	± 0.008	DESI DR2 BAO (sealed)

3 Pre-Registered Predictions

3.1 Primary Observable: S_8

The EFC prediction for the cosmic shear amplitude parameter is:

$$S_8^{\text{EFC}} \equiv \sigma_8 \left(\frac{\Omega_m}{0.3} \right)^{0.5} = 0.833 \times \left(\frac{0.310}{0.3} \right)^{0.5} \approx \mathbf{0.847} \pm 0.015. \quad (4)$$

This is to be compared against the Rubin DP2 measured value of S_8 . The Λ CDM expectation (from Planck 2018) is $S_8^{\Lambda\text{CDM}} \approx 0.832$. The EFC model predicts a modest upward shift:

$$\Delta S_8 \equiv S_8^{\text{EFC}} - S_8^{\Lambda\text{CDM}} \approx +0.015. \quad (5)$$

3.2 Angular Correlation Functions

The EFC model predicts an enhancement in $\xi_+(\theta)$ relative to Λ CDM at large angular scales. In the quasi-linear regime where lensing is dominated by the Weyl potential $(\Phi + \Psi)/2$, the ratio of EFC to Λ CDM correlation functions scales as Σ^2 to a good approximation. The predicted relative amplitude enhancement is:

$$\frac{\xi_+^{\text{EFC}}(\theta)}{\xi_+^{\Lambda\text{CDM}}(\theta)} \approx \Sigma^2(z_{\text{eff}}) \approx 1.08 \pm 0.04 \quad \text{for } \theta \in [20', 300']. \quad (6)$$

The effective lensing potential amplitude parameter Σ evaluated at the DP2 effective source redshift $z_{\text{eff}} \approx 0.65$ is:

$$\Sigma(z_{\text{eff}} = 0.65) = 1 + \alpha_\Sigma \mathcal{G}(a_{\text{eff}}) \approx 1.04 \pm 0.02, \quad (7)$$

yielding the lensing power spectrum enhancement $\Delta C_\ell/C_\ell \approx 8\%$ at $\ell \lesssim 300$.

3.3 Summary of Quantitative Predictions

Table 2: Pre-registered EFC predictions for Rubin DP2. Values are frozen as of the date of this document.

Observable	EFC Prediction	Λ CDM Expectation	Difference
S_8	0.847 ± 0.015	0.832 ± 0.013	+0.015
$\Sigma(z_{\text{eff}} = 0.65)$	1.04 ± 0.02	1.00	+4%
ξ_+ amplitude (large θ)	enhanced by $\sim 8\%$	baseline	+8%
$\mu(z_{\text{eff}} = 0.65)$	0.97 ± 0.03	1.00	−3%
$\eta(z = 0)$	1.10 ± 0.05	1.00	+10%

4 Falsification Criteria

The following constitutes the pre-registered falsification criterion for the KC4 lensing constraint:

1. **Primary criterion (KC4 falsification):**

If the Rubin DP2 measurement of $S_8 < 0.820$ at $> 2\sigma$ confidence, the KC4 lensing sector ($\eta \approx 1.10$) is ruled out.

2. **Shape-level criterion:**

If $\xi_+^{\text{obs}}(\theta)/\xi_+^{\Lambda\text{CDM}}(\theta) < 1.00$ at $\theta > 30'$ with $> 2\sigma$ significance, the large-scale lensing enhancement is excluded.

3. **Slip parameter criterion:**

If a joint measurement of (μ, Σ) from DP2 combined with spectroscopic data is inconsistent with $\eta > 1.05$ at $> 2\sigma$, the gravitational slip prediction is falsified.

4. **Tension criterion (model reassessment required):**

If $|S_8^{\text{obs}} - S_8^{\text{EFC}}| > 2\sigma$ in either direction, this document triggers a mandatory reassessment of the KC4 posterior constraints.

Note on S8 tension: The EFC model predicts $S_8 \approx 0.847$, which is *higher* than both ΛCDM and the currently observed tension direction. A Rubin DP2 measurement consistent with $S_8 \approx 0.76$ (the KiDS/HSC low- S_8 value) would falsify KC4 but would not falsify the broader EFC framework, as lower- η solutions remain available.

5 Analysis Protocol

5.1 Comparison Methodology

When Rubin DP2 data become available, the comparison shall be performed as follows:

1. Obtain the publicly released $\xi_{\pm}(\theta)$ or $C_{\ell}^{\kappa\kappa}$ two-point functions from the Rubin DP2 data release.
2. Compare against EFC theoretical predictions using the Regime-Consistent Measurement Principle (RCMP, DOI: 31222900), which requires that the angular scales used for comparison lie within the regime where the EFC modification is active ($z_{\text{eff}} < 2.0$, $\ell < 1000$).
3. Apply the same scale cuts and systematics treatment as the official Rubin DP2 analysis to avoid confirmation bias.
4. Report the χ^2 and $\Delta\chi^2$ relative to ΛCDM for the EFC prediction, using the Rubin DP2 covariance matrix.

5.2 Nuisance Parameters

The following nuisance parameters are *not* free in the EFC comparison; they are fixed to Rubin DP2 best-fit values as reported by the collaboration:

- Intrinsic alignment amplitude A_{IA}
- Photometric redshift bias Δz_i per tomographic bin
- Multiplicative shear bias m_i per tomographic bin
- Baryon feedback amplitude (fixed to Rubin DP2 value)

The EFC comparison varies only (μ_0, Σ_0) or equivalently η around the KC4 central value, holding all cosmological parameters fixed to the Rubin DP2 best-fit background cosmology. All EFC predictions are derived from a Boltzmann-consistent implementation of the EFC parameterisation (DOI: 31876324). No nuisance parameter freedom is used to improve agreement with data.

5.3 Scale Cuts

Angular scale cuts follow the RCMP (DOI: 31222900) regime criterion:

- ξ_{+} : use $\theta \in [0.5', 300']$ (all scales where EFC is active)
- ξ_{-} : use $\theta \in [4', 300']$
- $C_{\ell}^{\kappa\kappa}$: use $\ell \in [20, 1000]$

6 Reproducibility and Integrity

6.1 Source Documents

All EFC parameter values in this pre-registration are derived from the following independently archived and timestamped sources:

Table 3: Archived source documents for this pre-registration.

DOI (figshare)	Description
32010399	EFC Pre-Registered Predictions (parent document)
31876324	EFC Relativistic Action: Field Equations, μ , Σ , η
31990053	EFC Euclid DR1 Boltzmann Pre-Registration
31222900	Regime-Consistent Measurement Principle (RCMP)
31332730	$f\sigma_8$ Leave-One-Out Signal Analysis
31230703	Regime-Transition Fit to DESI DR2 BAO

All DOIs resolve via <https://doi.org/10.6084/m9.figshare.NNNNNNN>.

6.2 Sealed Posterior Hashes

The MCMC posterior runs underpinning the KC4 parameter values are sealed:

- `freeze_20260218_050713`: SHA-256 7a850cfa58477701... ($\alpha = -0.689$, $f\sigma_8(z=0.7) = 0.430$)
- `freeze_20260221_160857`: SHA-256 dbccda150abca0e7... ($\alpha = -0.702$)

These hashes verify that the $\eta \approx 1.10$ value used in this document was not adjusted after knowledge of the Rubin DP2 data.

6.3 Final Statement

This document constitutes a complete, frozen set of predictions for the EFC model applied to Rubin Observatory LSST DP2 cosmic shear data, prior to the availability of those data. All numerical values are derived from the archived sources listed above. No post-hoc modifications are permitted. The falsification criteria in Section 4 are binding.

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